


# BRAUER GROUPS

Report on second year coursework

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# Introduction

The following is an overview of my coursework that was part of my PhD program in the second year at TIFR,Mumbai. Here we are giving an overview of various properties of Brauer groups.

Throughout we have followed [Har77] for Algebraic geometric definitions and results. The first chapter is based on Brauer groups for fields where we know it is a torsion group. We mainly follow [CS21; GS17; FD93] for this section. We define the Serre residue and the Witt residues following [CS21]. Although we do not go into the details but we do need properties of Galois cohomology groups as per [Ser97].

In the second chapter we talk about Brauer groups for general schemes. There are two different notions in this case. The Brauer-Azumaya group is defined as isomorphism classes of Azumaya algebras and the cohomological Brauer group is $H^2_{\acute{e}t}(X, \mathbb{G}_m)$. We follow [Mil80] for the theory of Étale cohomology. Although for general schemes the cohomological Brauer group need not be torsion it turns out that for regular Noetherian schemes it is so. We also discuss Brauer groups for curves and how Brauer group of a scheme is related to that of its reduced subscheme. At the end we talk about the purity result in étale cohomology and that for Brauer groups where we follow [CS21; Mil80]. As a consequence we get Gysin exact sequence.

In the third chapter we state a result of Gabber that for any scheme $X$ with an ample invertible sheaf the Brauer-Azumaya group is isomorphic to the torsion part of the cohomological Brauer group. To understand this one views the elements of Brauer groups as gerbes. We have followed [Ols16] for the theory of algebraic stacks and gerbes. We define twisted sheaves and give a very brief idea of the proof.

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# Chapter 1

# Brauer Group of fields

[GS17; CS21; FD93] Fix a field $k$. A finite dimensional $k$-algebra $A$ is said to be a *central simple algebra* over $k$ if it has no non-trivial two sided ideals and its center is isomorphic to $k$. An easy example is the matrix algebras over $k$. For two such central simple $k$-algebras $A$ and $B$ we define them to be related if there are $n, m > 0$ such that

$$A \otimes_k M_m(k) \simeq B \otimes_k M_n(k).$$

This is an equivalence relation. Using the *Artin-Wedderburn* theorem this equivalence relation can be restated as follows. If $A \simeq M_n(D_1)$ and $B \simeq M_m(D_2)$ for $k$-division algebras $D_1, D_2$ then $A$ and $B$ are related if and only if $D_1$ and $D_2$ are isomorphic. The equivalence classes under this relation becomes (Abelian)group where tensor product is the binary operation. We denote it by $Br(k)$. The class of matrix algebras $[M_n(k)]$ is the identity element in the group. As there are no non-trivial division algebras over any field $k = \overline{k}$ we get $Br(k) = \{0\}$. More generally for $C_1$ fields (A field $k$ is said to be $C_1$ if for any homogeneous polynomial $f$ of degree $d$ in $n$ many variables with $d < n$ has a non-trivial solution.) the Brauer group is trivial. This result is due to Lang. A theorem of Tsen shows that for any field $k$ of transcendence degree 1 over an algebraically closed field is $C_1$.

Suppose $H^i(k, M)$ denotes the $i$th Galois cohomology group of a $Gal(k_s/k)$ module $M$. Then there is an equivalent description of $Br(k)$ which identifies $Br(k) = H^2(k, k_s^\times)$. This shows that $Br(k)$ is always torsion. It follows that $Br(\mathbb{R}) = \mathbb{Z}/2\mathbb{Z}$. For a field extension $K/k$ we have the restriction map $res : Br(k) \to Br(K)$ and if $[K : k] < \infty$ there is corestriction map $cor : Br(K) \to Br(k)$.

## 1.1 Residue maps

[CS21] A reference that I followed at times for residue maps and its relation to cup products etc. is the university lecture series [GMS03]. Let $R$ be a henselian discretely valued ring with field of fractions $K$ and residue field $k$. Denote the characteristic exponent of $k$ by $p$. So we have

$$K \subset K_{nr} \subset K_t \subset K_s$$

where $K_{nr}$ denotes the maximal unramified extension and $K_t$ denotes the maximal tamely ramified extension of $K$. Let $G = Gal(K_s/K), \Gamma = Gal(K_s/K_{nr}) \simeq Gal(k_s/k), I = Gal(K_s/K_{nr})$ and $N = Gal(K_t/K_{nr})$ the tame inertia group. For a continuous discrete torsion $\Gamma$-module with trivial $p$-torsion, using the Hochschild-Serre spectral sequence we obtain split exact sequences for all $i \geq 1$

$$0 \to H^i(k, C) \to H^i(K, C) \xrightarrow{r} H^{i-1}(k, C(-1)) \to 0$$

where $C(-1) = Hom(N, C)$. The map $r$ is called the **Serre Residue**.

**Remark 1.** *If $C = \mu_n$ with $(n, p) = 1$ and $i = 1$ the Serre residue*

$$r : H^1(K, \mu_n) \to \mathbb{Z}/n\mathbb{Z}$$

*is induced by the valuation map $\nu : K^\times \to \mathbb{Z}$ if one identifies $H^1(K, \mu_n) \simeq K^\times/(K^\times)^n$ using Kummer theory.*

If moreover the residue field $k$ is perfect using the Hochschild-Serre spectral sequence

$$H^p(\Gamma, H^q(K_{nr}, K_s^\times)) \implies H^{p+q}(K, K_s^\times)$$

and the Hilbert's theorem 90, one gets $H^2(\Gamma, K_{nr}^\times) \xrightarrow{\simeq} Br(K)$. Applying the valuation $\nu$ and the isomorphism $H^2(\Gamma, \mathbb{Z}) \simeq H^1(\Gamma, \mathbb{Q}/\mathbb{Z})$, we get the **Witt Residue map**.

$$r_W : Br(K) \to H^1(k, \mathbb{Q}/\mathbb{Z}).$$

**Remark 2.** *The two residues defined above are related as follows. For a henselian dvr $R$ let $n$ be a positive integer invertible on $R$. The composition*

$$H^2(K, \mu_n) \xrightarrow{\simeq} Br(K)[n] \xrightarrow{r_W} H^1(k, \mathbb{Z}/n\mathbb{Z}).$$

*is equal to the negative of the Serre residue $r : H^2(K, \mu_n) \to H^1(k, \mathbb{Z}/n\mathbb{Z})$.*

*One uses the spectral sequence and Yoneda pairings and extension of groups to prove it. Theory about the Yoneda pairing can be found in [HS97].*

## 1.2 The Faddeev exact sequences

Let $k$ be a perfect field with algebraic closure $\overline{k} = k_s$ and Galois group $\Gamma_k = Gal(\overline{k}/k)$. To any monic irreducible polynomial $P(t) \in k[t]$ consider $\mathbb{Z}_P$ the free abelian group generated by the roots of $P(t)$ in $\overline{k}$ with its natural action by $\Gamma$. The natural exact sequence of $\Gamma$-modules

$$0 \to \overline{k}^{\times} \to \overline{k}(t)^{\times} \to Div(\mathbb{A}^1_{\overline{k}}) \to 0$$

and

$$0 \to \overline{k}^{\times} \to \overline{k}(t)^{\times} \to Div(\mathbb{P}^1_{\overline{k}}) \to \mathbb{Z} \to 0$$

using spectral sequence gives rise to the following **Faddeev exact sequences**.

**Theorem 1.** *(Faddeev) There exists split exact sequences*

$$0 \to Br(k) \to Br(k(t)) \to \bigoplus Hom(\Gamma_{k(P)}, \mathbb{Q}/\mathbb{Z}) \to 0$$

*where the direct sum runs over all monic irreducible polynomials $P(t) \in k[t]$ and $k(P) = k[t]/(P(t))$. Moreover for each $P(t)$, the map $Br(k(t)) \to Hom(\Gamma_{k(P)}, \mathbb{Q}/\mathbb{Z})$ is the Witt residue map corresponding to the valuation for which $P(t)$ is a uniformizer.*

*Also there is an exact sequence*

$$0 \to Br(k) \to Br(k(t)) \to \bigoplus_{x \in (\mathbb{P}^1_k)^{(1)}} H^1(k(x), \mathbb{Q}/\mathbb{Z}) \to H^1(k, \mathbb{Q}/\mathbb{Z}) \to 0.$$

*where the third arrows are each Witt residue and the fourth arrow is the sum of corestriction maps $cores_{k(x)/k}$ for each closed points $x$ of $\mathbb{P}^1_k$.*

# Chapter 2

# Brauer groups of schemes

There are two ways to generalize the Brauer groups of fields to that of schemes. One way is to consider the isomorphism classes of Azumaya algebras over a scheme instead of the central simple algebras over a field. Equivalently the cohomological description $H^2(k, k_s^\times)$ of $Br(k)$ can be generalized by means of Étale cohomology on a scheme. Étale cohomology was introduced by A.Grothendieck.

A category $C$ consisting with a class of morphisms $E$ that contains all the isomorphisms and is closed under base change and taking composition is called a site. We will mostly consider the big and small étale sites over a base scheme $X$. The former means taking all schemes over $X$ with the class of morphisms being étale and the later means taking all schemes over $X$ with the structure map being étale.

## 2.1 The Brauer-Azumaya group

[CS21; Mil80]Let $X$ be a scheme. Then for a sheaf of $\mathcal{O}_X$-algebras say $\mathcal{A}$ the following are equivalent.

i) Each $\mathcal{A}_x$ is an Azumaya algebra over the local ring $\mathcal{O}_{X,x}$ i.e. $\mathcal{A}_x$ is a free module of finite rank and $\mathcal{A}_x \otimes \mathcal{A}_x^{op} \simeq M_n(\mathcal{O}_{X,x})$ for some $n \in \mathbb{N}$.

ii) $\mathcal{A}(x) = \mathcal{A}_x \otimes_{\mathcal{O}_{X,x}} k(x)$ is a central simple algebra over $k(x)$.

iii) There is an etale covering $\{U_i \to X\}$ such that $\mathcal{A} \otimes_{\mathcal{O}_X} \mathcal{O}_{U_i} \simeq M_{r_i}(\mathcal{O}_{U_i})$.

If one of the above equivalent conditions are satisfies we call it an **Azumaya algebra** over $X$.

For two such Azumaya algebras say $\mathcal{A}$ and $\mathcal{B}$ we call them to be equivalent if there exists locally free sheaves of finite ranks say $\mathcal{P}$ and $\mathcal{Q}$ on $X$ such that

$$\mathcal{A} \otimes_{\mathcal{O}_X} \underline{\mathcal{E}nd}(\mathcal{P}) \simeq \mathcal{B} \otimes_{\mathcal{O}_X} \underline{\mathcal{E}nd}(\mathcal{Q}).$$

These isomorphism classes under tensor product defines the **Brauer-Azumaya group** of the scheme $X$ denoted by $Br_{Az}(X)$.

## 2.2 The Brauer-Grothendieck group

[CS21]Grothendieck defined the cohomological Brauer group as $Br(X) = H^2_{\acute{e}t}(X, \mathbb{G}_m)$. Given a morphism $f : X \to Y$ there exists the induced map $f^* : Br(Y) \to Br(X)$. If $X =$Spec$k$ for a field $k$ it is same as $H^2(k, k_s^\times)$. In particular a torsion group. However in general this may not be true. We will provide an example at the end.

### 2.2.1 Relation with the reduced subscheme

Let $X$ be noetherian scheme with its reduced subscheme $X_{red}$. For some $n \in \mathbb{N}$ there exists the chain of closed immersions

$$X_{red} = X_0 \subset X_1 \subset .. \subset X_n = X$$

with their ideal sheaves $\mathcal{I}_j \subset \mathcal{O}_{X_j}$ such that $\mathcal{I}_j^2 = 0$ and $\mathcal{O}_{X_{(j-1)}} = \mathcal{O}_{X_j}/\mathcal{I}_j$. On each $X_j$

$$0 \to \mathcal{I}_j \to \mathbb{G}_{m,X_j} \to r_*\mathbb{G}_{m,X_{j-1}} \to 0$$

where $r : X_{j-1} \to X_j$ is the closed immersion and the second arrow is given by $x \mapsto 1 + x$. Using suitable spectral sequences on the above exact sequence and doing it for each $j$ in the previous chain of closed subschemes one obtain the following relations between $Br(X)$ and $Br(X_{red})$.

**Theorem 2.** *i) If $X$ is affine or if dim$X \leq 1$ then $Br(X) \to Br(X_{red})$ is an isomorphism.*
*ii) If dim$X \leq 2$ then $Br(X) \to Br(X_{red})$ is surjective.*

### 2.2.2 Going over the generic point

Here our scheme $X$ will be noetherian and integral with the generic point $j$ :spec$F \to X$. $X$ will be called geometrically locally factorial if for each $x \in X$ the local rings in étale topology $\mathcal{O}^{sh}_{X,x}$ is unique factorization domain. A regular local ring is geometrically factorial. So if moreover we assume $X$ to be geometrically locally factorial then there is the exact sequence

$$0 \to \mathbb{G}_{m,X} \to j_*\mathbb{G}_{m,F} \to \bigoplus_{D \in X^{(1)}} i_{D_*}\mathbb{Z}_{k(D)} \to 0$$

Taking cohomology it follows that $Br(X) \hookrightarrow Br(F)$. In particular that $Br(X)$ is torsion. More generally for any non empty open set $U \subset X$ the induced map $Br(X) \to Br(U)$ is injective.

### 2.2.3 Brauer group for curves

A curve will mean a scheme of dimension one. By the previous section if the curve $X$ is integral and regular then $Br(X)$ is torsion.

**Theorem 3.** *Let $X$ be an integral regular scheme of dimension one with the generic point $specF$. Then*

*i) For each prime $l$ that is invertible on $X$ there is an exact sequence*

$$.. \to H^i_{\acute{e}t}(X, \mathbb{G}_m)\{l\} \to H^i_{\acute{e}t}(F, \mathbb{G}_m)\{l\} \to \bigoplus_{x \in X^{(1)}} H^{i-1}(k(x), \mathbb{Q}_l/\mathbb{Z}_l) \to H^{i+1}_{\acute{e}t}(X, \mathbb{G}_m)\{l\} \to ..$$

*starting from $i = 2$.*

*ii) If for each closed point $x \in X$ the residue field $k(x)$ is perfect then*

$$.. \to H^i_{\acute{e}t}(X, \mathbb{G}_m) \to H^i_{\acute{e}t}(F, \mathbb{G}_m) \to \bigoplus_{x \in X^{(1)}} H^{i-1}(k(x), \mathbb{Q}/\mathbb{Z}) \to H^{i+1}_{\acute{e}t}(X, \mathbb{G}_m) \to ..$$

*starting from $i = 2$. Moreover for each $x \in X^{(1)}$ the maps $Br(F) \to H^1(k(x), \mathbb{Q}/\mathbb{Z})$ is the composition of of $Br(F) \to Br(F^{sh}_x)$ where $F^{sh}_x$ is the field of fractions of the henselization of the local ring $\mathcal{O}_{X,x}$ and the Witt residue $Br(F^{sh}_x) \to H^1(k(x), \mathbb{Q}/\mathbb{Z})$.*

If $X =$Spec$R$ where $R$ is an henselian discrete valued ring with field of fractions $K$ and perfect residue field $k$, from the surjectivity of Witt residue the following exact sequence follows.

$$0 \to Br(R) \to Br(K) \to H^1(k, \mathbb{Q}/\mathbb{Z}) \to 0.$$

Now the map $R \to k$ induces $Br(k) \xrightarrow{\sim} Br(R)$. In particular if $K$ is a local field then $Br(K) \simeq H^1(\hat{\mathbb{Z}}, \mathbb{Q}/\mathbb{Z}) \simeq \mathbb{Q}/\mathbb{Z}$.

In general if $X$ is a Noetherian curve(not necessarily regular) singular then also $Br(X)$ is torsion. Consider the points of dimension one $y_1, .., y_n$ with $i_{y_i}$ :Spec $k(y_i) \hookrightarrow X$ and let we denote the closed points by $x =$Spec$k(x) \hookrightarrow X$. Then there is an exact sequence

$$0 \to \mathbb{G}_{m,X} \to \bigoplus_{j=1}^{n} i_{y_j *} \mathbb{G}_{m,k(y_j)} \to \bigoplus_x i_{x*} F_x \to 0.$$

Taking cohomology it follows that $Br(X)$ is torsion. Applying cohomology and Leray spectral sequences we get that

$$0 \to \bigoplus_x H^1(k(x), F_x) \to Br(X) \to \bigoplus_{j=1}^{n} Br(k(y_j))$$

which implies that $Br(X)$ is torsion. □

### 2.2.4 Purity for Brauer groups

[Mil80; CS21] Let $X$ be a scheme with $i : Z \hookrightarrow X$ a closed subscheme and $U$ be the open immersion $j$ of the compliment $X \setminus Z$. Then there are six functors as following :

$$i^* : Sh_{ét}(X) \to Sh_{ét}(Z) \qquad j_! : Sh_{ét}(U) \to Sh_{ét}(X)$$
$$i_* : Sh_{ét}(Z) \to Sh_{ét}(X) \qquad j^* : Sh_{ét}(X) \to Sh_{ét}(U)$$
$$i^! : Sh_{ét}(X) \to Sh_{ét}(Z) \qquad j_* : Sh_{ét}(U) \to Sh_{ét}(X)$$

Here each functor is left adjoint to the one listed below. They are related by the exact sequence

$$0 \to j_! j^* \mathcal{F} \to \mathcal{F} \to i_* i^* \mathcal{F} \to 0$$

which induces the long exact sequence

$$.. \to H^i_Z(X, \mathcal{F}) \to H^i_{ét}(X, \mathcal{F}) \to H^i_{ét}(U, \mathcal{F}|U) \to H^{i+1}_Z(X, \mathcal{F}) \to ..$$

Here $H^i_Z(X, -)$ is the right derived functor of the left exact functor $\Gamma(X, i^!-)$. Note for any sheaf $\mathcal{F}$ on $X$ we have

$$\Gamma(Z, i^! \mathcal{F}) = Ker(\mathcal{F}(X) \to \mathcal{F}(U)).$$

Details about derived category can be found in [Wei94].

**Theorem 4.** *(Gabber) Let $X$ be a regular scheme, $i : Z \hookrightarrow X$ be a closed regular subscheme of codimension $c$ everywhere. Let $l$ be a prime different from the residual characteristics of $X$ and $m$ be a positive integer. Suppose $\mathcal{D}^+(Z_{et})$ denotes the bounded derived category of étale sheaves on $Z$. Then in $\mathcal{D}^+(Z_{et}, \mathbb{Z}/l^m)$ we have an isomorphism $\mathbb{Z}/l^m \xrightarrow{\sim} \mathbf{R}i^!(\mathbb{Z}/l^m)(c)[2c]$. That is in particular*

$$\underline{H}^n_Z(X, \mathbb{Z}/l^m) = 0 (n \neq 2c), \qquad \underline{H}^{2c}_Z(X, \mathbb{Z}/l^m) = \mathbb{Z}/l^m(-c)_Z.$$

*where $(\mathbb{Z}/l^m)(0)_X = (\mathbb{Z}/l^m)$, for $c > 0$ $(\mathbb{Z}/l^m)(c)_X = \mu_{n,X}^{\otimes c}$ and for $c < 0$ $(\mathbb{Z}/l^m)(c)$ denotes the sheaf associated to the presheaf that sends any $Y \to X$ étale to $Hom_Y((\mathbb{Z}/l^m)(-c), (\mathbb{Z}/l^m)_Y)$.*

As a consequence we obtain the **Gysin exact sequence**

$$.. \to H^{i-2c}_{ét}(Z, \mathbb{Z}/l^m(-c) \to H^i_{ét}(X, \mathbb{Z}/l^m) \to H^i_{ét}(U, \mathbb{Z}/l^m) \to H^{i-2c+1}_{ét}(Z, \mathbb{Z}/l^m(-c)) \to ..$$

In the special case if $X = \mathrm{Spec} R$ with $R$ an henselian discretely valued ring with residue field $k$ and the field of fraction $K$ this induces

$$H^i(k, \mu_{l^m}) \to H^i(K, \mu_{l^m}) \xrightarrow{\partial_R} H^{i-1}(k, \mathbb{Z}/l^m) \to H^{i+1}(k, \mu_{l^m}) \to ..$$

The map $\partial_R$ is called the **Gysin residue**.

**Theorem 5.** *The Gysin residue map defined above is the negative of the Serre residue defined in 1.1.*

Next we will state purity results for Brauer groups.

**Theorem 6.** *Let $X$ be a regular integral Noetherian scheme, let $Z \subset X$ be a regular closed subset of pure codimension $c$. Let $U \subset X$ be the open set $X \setminus Z$. For a prime $l$ invertible on $X$*

*i) If $c \geq 1$, then the restriction map $Br(X)\{l\} \to Br(U)\{l\}$ is an isomorphism.*

*ii) If $c = 1$ and $D_1, .., D_m$ are the connected components of $Z$, then*

$$0 \to Br(X)\{l\} \to Br(U)\{l\} \to \bigoplus_{i=1}^{m} H^1_{et}(D_i, \mathbb{Q}_l/\mathbb{Z}_l).$$

**Theorem 7.** *Let $X$ be an excellent, regular, integral,Noetherian scheme. Let $U \subset X$ be a dense open subset of codimension at least 2. Then for any prime $l$ that is invertible on $X$ the restriction map*

$$Br(X)\{l\} \to Br(U)\{l\}$$

*is an isomorphism.*

The following theorem which generalizes the above was proved by Česnavičius.

**Theorem 8.** *Let $X$ be a regular, integral, Noetherian scheme. Let $U \subset X$ be an open subset whose complement is of codimension at least 2. Then the restriction map $Br(X) \to Br(U)$ is an isomorphism.*

## 2.3 Example of a non torsion Brauer group

It is clear that such an example must exist in dimension at least 2. Consider a smooth projective variety $Y$ over a field $k$ of char 0. Define $X = Y \times k[\epsilon]$ such that $\epsilon^2 = 0$. Then $X_{red} = Y$ and let $i : Y \to X$ be the closed immerison. It has a section which we call $s : X \to Y$ i.e. $s \circ i = id$.

The following split exact sequence

$$0 \to i_*\mathcal{O}_Y \to \mathbb{G}_{m,X} \to i_*\mathbb{G}_{m,Y} \to 0.$$

induces the following short exact sequence

$$0 \to H^2_{\acute{e}t}(Y, \mathcal{O}_Y) \to Br(X) \to Br(Y) \to 0.$$

If we consider $Y = C \times C'$ where both $C, C'$ are curves of genus $g$. Hence $H^2(Y, \mathcal{O}_Y)$ is nonzero i.e. a vector space over $k$. It proves existence of non-torsion elements in $Br(X)$. Note this also shows that for schemes of dimension more than one the natural map $Br(X) \to Br(X_{red})$ need not be injective. □

# Chapter 3

# Comparison of the two Brauer groups

We will need the language of algebraic stacks and gerbes for which we are following [Ols16]. Content of this section are from [CS21](ChIV). In the previous section we talked about the Brauer-Azumaya groups and the cohomological Brauer group.

**Proposition 1.** *Let $X$ be a scheme. Then the order $[\mathcal{A}]$ class represented by an Azumaya algebra of degree $n$ divides $n$. If $X$ is quasi compact or has finitely many connected components, then $Br_{Az}(X)$ is torsion.*

There is a well defined map from $Br_{Az}(X) \to Br(X)$ which is injective. However the surjectivity is not always there. Before defining the map note that we may view the elements of $Br(X)$ as gerbes over $X$ banded by $\mathbb{G}_m$. Roughly speaking a gerbe over a site $C$ is a stack $p : F \to C$ such that sections exist locally and two sections must be locally isomorphic. A gerbe such that for each point $x$ the sheaf $\underline{Aut}_x$ is isomorphic to $\mathbb{G}_{m,x}$ is said to be banded by $\mathbb{G}_m$.[Mil80] For an Azumaya algebra $\mathcal{A}$ on $X$ consider the stack $\mathcal{G}_{\mathcal{A}}$ on $X$ given by the triple

$$\{(T, E, \sigma) | T \xrightarrow{f} X, E \text{ locally free sheaf on } T, \underline{\mathcal{E}nd}(E) \underset{\sigma}{\xrightarrow{\sim}} f^*\mathcal{A}\}.$$

For two objects $t' = (T', E', \sigma'), t := (T, E, \sigma)$ morphisms $t' \to t$ are pairs $(f, f^b)$ such that $f : T' \to T$ morphism of $S$-schemes and $f^b : f^*E \xrightarrow{\sim} E'$ as locally free sheaves on $T'$ such that the diagram below commutes.

$$\begin{array}{ccc} \underline{\mathcal{E}nd}(f^*E) & \xrightarrow{\hat{f^b}} & \underline{\mathcal{E}nd}(E') \\ \downarrow \sim & & \downarrow \sigma' \\ f^*\underline{\mathcal{E}nd}(E) & \xrightarrow{f^*\sigma} & f^*\mathcal{A} \end{array}$$

It follows that for any object $t = (T, E, \sigma)$ the sheaf $\underline{Aut}_t \simeq \mathbb{G}_{m,t}$. So we have a map

$$\varphi : Br_{Az}(X) \to Br(X)$$
$$[\mathcal{A}] \mapsto \mathcal{G}_{\mathcal{A}}.$$

It is immediate that this is an injection.

**Theorem 9.** *(Gabber) Suppose $X$ be a scheme with an ample invertible sheaf $\mathcal{L}$. Then the map $Br_{Az}(X) \to Br(X)_{tors}$ is an isomorphism.*

The proof uses the notion of twisted sheaves on a gerbe. A locally free sheaf $\mathcal{E}$ on a $\mathbb{G}_m$ gerbe $\mathcal{G}$ is said to be $n$-twisted if for any $x$ :Spec$k \to \mathcal{G}$ the $k$-vector space $x^*\mathcal{E}$ is being acted by $\mathbb{G}_{m,k}$ via

$$\lambda \cdot x = \lambda^n x.$$

There is another notion of $\alpha$-twisted sheaves on scheme $X$ which can be found in the thesis of A. Căldăraru. For $\alpha \in H^2(X, \mathbb{G}_m)$ if there is an étale covering $\{U_i \to X\}$ trivializing $\alpha$ with respect to which there is čech 2-cocycle $\alpha_{ijk} \in \Gamma(U_{ijk}, \mathbb{G}_m)$ then an $\alpha$-twisted sheaf is a collection $\mathcal{M}_i$ of coherent sheaves on $U_i$ together with

$$\varphi_{ij} : \mathcal{M}_j \otimes \mathcal{O}_{U_{ij}} \xrightarrow{\sim} \mathcal{M}_i \otimes \mathcal{O}_{U_{ij}}$$

such that over $U_{ijk}$

$$\varphi_{ij} \circ \varphi_{jk} = \alpha_{ijk}\varphi_{ik}.$$

It turns out that for $\alpha \in H^2(X, \mathbb{G}_m)$ the category of 1-twisted sheaves on the associated gerbe $\mathcal{X}_\alpha$ is equivalent to the category of $\alpha$-twisted sheaves on $X$. The gerbe $\mathcal{G}_{\mathcal{A}}$ has a tautological locally free 1-twisted sheaf. Conversely also if $\pi : \mathcal{G} \to S$ is gerbe and $\mathcal{F}$ be a locally free 1-twisted sheaf on the gerbe $\mathcal{G}$ then $\mathcal{A} := \pi_* \underline{\mathcal{E}nd}(\mathcal{F})$ is an Azumaya algebra on $S$ such that $\pi^*\mathcal{A} \simeq \underline{\mathcal{E}nd}(\mathcal{F})$ and $\mathcal{G} \simeq \mathcal{G}_{\mathcal{A}}$.

We are asking about the surjectivity of the map $\varphi$ above and from the earlier discussion it is sufficient to find a locally free 1-twisted sheaf on a given gerbe on $S$ corresponding to a class $\alpha \in H^2(S, \mathbb{G}_m)$. It uses a result of Gabber for the affine case to get an $\alpha$-twisted sheaf $F_s$ on $X_s$ for some $s \in \Gamma(X, \mathcal{L}^{\otimes n})$. A proof can be found here.

# Bibliography


[CS21] Jean-Louis Colliot-Thélène and Alexei N. Skorobogatov. *The Brauer-Grothendieck group*. Vol. 71. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. A Series of Modern Surveys in Mathematics [Results in Mathematics and Related Areas. 3rd Series. A Series of Modern Surveys in Mathematics]. Springer, Cham, [2021] ©2021, pp. xv+453. ISBN: 978-3-030-74247-8; 978-3-030-74248-5. DOI: 10.1007/978-3-030-74248-5. URL: https://doi.org/10.1007/978-3-030-74248-5.

[FD93] Benson Farb and R. Keith Dennis. *Noncommutative algebra*. Vol. 144. Graduate Texts in Mathematics. Springer-Verlag, New York, 1993, pp. xiv+223. ISBN: 0-387-94057-X. DOI: 10.1007/978-1-4612-0889-1. URL: https://doi.org/10.1007/978-1-4612-0889-1.

[GMS03] Skip Garibaldi, Alexander Merkurjev, and Jean-Pierre Serre. *Cohomological invariants in Galois cohomology*. Vol. 28. University Lecture Series. American Mathematical Society, Providence, RI, 2003, pp. viii+168. ISBN: 0-8218-3287-5. DOI: 10.1090/ulect/028. URL: https://doi.org/10.1090/ulect/028.

[GS17] Philippe Gille and Tamás Szamuely. *Central simple algebras and Galois cohomology*. Second. Vol. 165. Cambridge Studies in Advanced Mathematics. Cambridge University Press, Cambridge, 2017, pp. xi+417. ISBN: 978-1-316-60988-0; 978-1-107-15637-1.

[Har77] Robin Hartshorne. *Algebraic geometry*. Vol. No. 52. Graduate Texts in Mathematics. Springer-Verlag, New York-Heidelberg, 1977, pp. xvi+496. ISBN: 0-387-90244-9.

[HS97] P. J. Hilton and U. Stammbach. *A course in homological algebra*. Second. Vol. 4. Graduate Texts in Mathematics. Springer-Verlag, New York, 1997, pp. xii+364. ISBN: 0-387-94823-6. DOI: 10.1007/978-1-4419-8566-8. URL: https://doi.org/10.1007/978-1-4419-8566-8.

[Mil80] James S. Milne. *Étale cohomology*. Vol. No. 33. Princeton Mathematical Series. Princeton University Press, Princeton, NJ, 1980, pp. xiii+323. ISBN: 0-691-08238-3.

[Ols16] Martin Olsson. *Algebraic spaces and stacks*. Vol. 62. American Mathematical Society Colloquium Publications. American Mathematical Society, Providence, RI, 2016, pp. xi+298. ISBN: 978-1-4704-2798-6. DOI: 10.1090/coll/062. URL: https://doi.org/10.1090/coll/062.

[Ser97] Jean-Pierre Serre. *Galois cohomology*. Translated from the French by Patrick Ion and revised by the author. Springer-Verlag, Berlin, 1997, pp. x+210. ISBN: 3-540-61990-9. DOI: 10.1007/978-3-642-59141-9. URL: https://doi.org/10.1007/978-3-642-59141-9.

[Wei94] Charles A. Weibel. *An introduction to homological algebra*. Vol. 38. Cambridge Studies in Advanced Mathematics. Cambridge University Press, Cambridge, 1994, pp. xiv+450. ISBN: 0-521-43500-5; 0-521-55987-1. DOI: 10.1017/CBO9781139644136. URL: https://doi.org/10.1017/CBO9781139644136.